MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYETHYLENE TEREPHTHALATE, NON-CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: -60°C (-76°F) to +135°C (+275°F).

Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2.1):

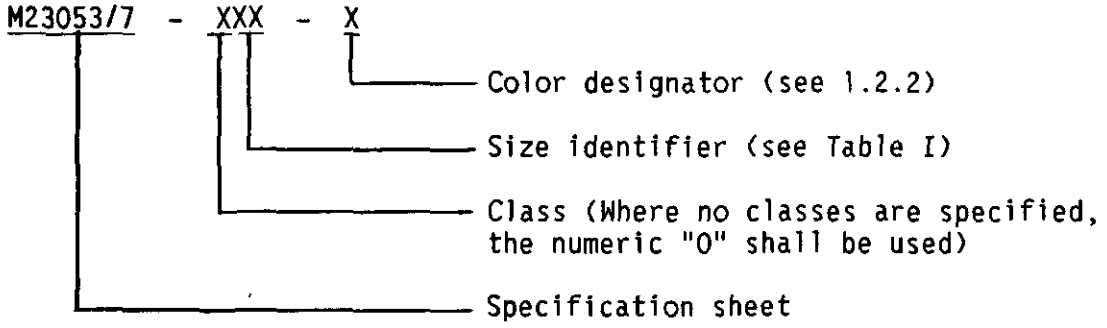
Class 1 - Single longitudinal seam, thermally bonded
Class 2 - Spirally wound, resin bonded

Color: The heat shrinkable sleeving shall be furnished in clear, as manufactured. The acquiring activity may specify colors for Class 2 sleeving only (see 6.2.1). Colors shall conform to Class I of MIL-STD-104.

Longitudinal change:

Class 1: 30 percent maximum
Class 2: 35 percent maximum

Military part number: The military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



NOTE: Entire specification sheet revised.

FSC 5970

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Example: Class 1, .125 (3.2 mm), as supplied ID sleeving shall be identified as M23053/7-103-C.

TABLE I. Construction details, inches (mm). 1/

Military part number 1/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/7-101-C	No longer	manufactured	
M23053/7-102-C	No longer	manufactured	
M23053/7-103-C	.125(3.2)	.088(2.2)	.0027 ± .0004(.0687 ± .010)
M23053/7-104-C	.187(4.8)	.131(3.3)	.0027 ± .0004(.0687 ± .010)
M23053/7-105-C	.250(6.4)	.175(4.6)	.0027 ± .0004(.0687 ± .010)
M23053/7-106-C	.375(9.5)	.263(6.7)	.0027 ± .0004(.0687 ± .010)
M23053/7-107-C	.500(12.7)	.350(8.9)	.0027 ± .0004(.0687 ± .010)
M23053/7-108-C	.750(19.1)	.525(13.4)	.0027 ± .0004(.0687 ± .010)
M23053/7-109-C	1.000(25.4)	.700(17.8)	.0027 ± .0004(.0687 ± .010)
M23053/7-110-C	1.375(34.9)	.963(24.5)	.0027 ± .0004(.0687 ± .010)
M23053/7-111-C	1.500(38.1)	1.050(26.7)	.0027 ± .0004(.0687 ± .010)
M23053/7-112-C	2.000(50.8)	1.400(35.6)	.0027 ± .0004(.0687 ± .010)
M23053/7-113-C	2.500(63.5)	1.750(44.5)	.0027 ± .0004(.0687 ± .010)
M23053/7-114-C	3.000(76.2)	2.100(53.3)	.0027 ± .0004(.0687 ± .010)
M23053/7-115-C	4.000(101.6)	2.800(71.1)	.0027 ± .0004(.0687 ± .010)
Class 2			
M23053/7-201-*	.015(.38)	.011(.28)	
M23053/7-202-*	.020(.50)	.015(.38)	
M23053/7-203-*	.025(.63)	.019(.48)	
M23053/7-204-*	.030(.76)	.023(.58)	
M23053/7-205-*	.042(1.06)	.032(.81)	3/
M23053/7-206-*	.046(1.17)	.035(.89)	
M23053/7-207-*	.062(1.57)	.047(1.19)	
M23053/7-208-*	.093(2.36)	.070(1.78)	
M23053/7-209-*	.125(3.18)	.094(2.4)	
M23053/7-210-*	.187(4.75)	.140(3.6)	
M23053/7-211-*	.250(6.35)	.188(4.8)	
M23053/7-212-*	.312(7.92)	.234(5.9)	
M23053/7-213-*	.375(9.53)	.281(8.1)	
M23053/7-214-*	.500(12.7)	.375(9.5)	
M23053/7-215-*	.625(15.88)	.469(11.9)	
M23053/7-216-*	.750(19.1)	.563(14.3)	
M23053/7-217-*	1.000(25.4)	.750(19.1)	
M23053/7-218-*	1.125(28.58)	.844(21.4)	3/
M23053/7-219-*	1.250(31.75)	.938(23.8)	
M23053/7-220-*	1.375(34.93)	1.032(26.2)	
M23053/7-221-*	1.500(38.1)	1.125(28.6)	
M23053/7-222-*	1.750(44.45)	1.313(33.4)	
M23053/7-223-*	2.000(50.8)	1.500(38.1)	

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TABLE I. Construction details, inches (mm). 1/ - Continued

1/ Diameter limits for the object to be enclosed shall be no less than 85 percent of the as supplied diameter.

2/ Class 1 wall thickness measurements shall be made away from the seam.

3/ Because of irregular shrinkdown of Class 2 sleevings, wall thickness shall be determined "as supplied" as follows:

Nominal ID (as supplied)		Wall thickness range (as supplied)	
inch	(mm)	inch	(mm)
.015 to .061	(.381 to 1.55)	.0015 to .004	(.038 to .102)
.062 to .074	(1.57 to 1.88)	.0015 to .005	(.038 to .127)
.075 to .094	(1.91 to 2.39)	.0015 to .006	(.038 to .152)
.095 to .124	(2.41 to 3.15)	.0015 to .008	(.038 to .203)
.125 to 2.000	(3.18 to 50.8)	.0015 to .010	(.033 to .254)

The acquiring activity (see 6.2.1) may specify the exact wall thickness required within each range.

Wall thickness tolerance (as supplied)

Wall		Tolerance	
inch	(mm)	inch	(mm)
.0015	(.038)	± .0005	(.0127)
.0016 to .003	(.041 to .076)	± .0008	(.0203)
.0031 to .005	(.079 to .127)	± .0010	(.0254)
.0051 to .010	(.130 to .254)	± .0015	(.0381)

4/ The asterisk in the part number shall be replaced by color code designations (see 1.2.2).

Unrestricted shrinkage: Test method 4.6.2.2; 160° ± 2°C (320° ± 4°F) for 5 minutes.

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TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Restricted shrinkage	No cracks	4.6.6.1.1 160° ± 2°C (320° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Color stability, Class 2	Pass	4.6.16 150° ± 2°C (302° ± 4°F), 48 hours
Clarity stability, clear sleeving	Pass	4.6.17 150° ± 2°C (302° ± 4°F), 24 hours
Heat shock	No cracks, flowing or dripping	4.6.8 180° ± 2°C (356° ± 4°F)
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Seam strength, Class 1, PPI (Kg/m), minimum	25 (450)	4.6.2 ASTM D 882, Method A
Tensile strength, psi (MPa), minimum	20,000 (138)	4.6.14 ASTM D 638, 20 inches/minute
Ultimate elongation, percent, minimum	50	4.6.14 ASTM D 638, 20 inches/minute
Dielectric strength, volts/mil (Kv/mm) minimum	2,500 (98.4)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm, minimum	10^{14}	4.6.2 ASTM D 876
Specific gravity, minimum	1.3	4.6.2 ASTM D 792
Corrosion	No corrosion	4.6.10.1 and 4.5.10.2 155 ± 2°C (311° ± 4°F), 16 hours

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Low temperature flexibility	No cracking	4.6.7.1 -60° ± 1°C (-76° ± 2°F)
Water absorption, percent, maximum	0.8	4.6.2 ASTM D 570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.8 150° ± 2°C (302° ± 4°F), 336 hours
Tensile strength, psi (MPa), minimum	Class 1 - 18,000 (124) Class 2 - 13,000 (89.7)	4.6.14 ASTM D 638, 20 inches/min
Ultimate elongation, percent, minumum	Class 1 - 55 Class 2 - 40	4.6.14 ASTM D 638, 20 inches/min
Dielectric strength volt/mil (Kv/mm), minimum	2,500 (63.5)	4.6.2 ASTM D 876
Fluid resistance, properties after:		4.6.11
Tensile strength psi (MPa), minimum	20,000 (98.4)	
Dielectric strength, volt/mil (Kv/mm), minimum	2,500(63.5)	
Flammability	Self-extinguishing 1 minute, 25 percent maximum indicator flag burned or charred	4.6.15.1
Fungus resistance	No growth	4.6.2 ASTM G 21

1/ Unless otherwise specified, the requirements, test procedures and conditions are for all classes.

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Shelf life: Supplier shall certify to storage at 40°C (104°F) for 4 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: To provide a tight protective covering for relays, electronic coils and components and small capacitor cans.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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